ELITE IGCSE ACADEMY

Private Past Paper Solutions

Jan 2020 P1H Worked Solutions

Jan 2020 | P1H

25 questions ▪ question image followed by worked solution

PREPARED BY

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TEACHER EDITION

Worked answers included

PRIVATE

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PAST PAPER QUESTION

Q#: 1 Marks: 4

TOPIC: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

1 The point A has coordinates $(5, -4)$
The point B has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of AB.

(.......................... , ...)
(2)

Line **L** has equation $y = 2 - 3x$

(b) Write down the gradient of line **L**.

....
(1)

Line **L** has equation $y = 2 - 3x$

(c) Does the point with coordinates $(100, -302)$ lie on line **L**?
You must give a reason for your answer.

....

....

....
(1)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Coordinate geometry. The tag is correct.

Solution

(a) The midpoint of $A(5, -4)$ and $B(13, 1)$ is

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

$$= (9, -1.5)$$

(b)

$$y = 2 - 3x$$

The gradient is -3.

(c) For $x = 100$,

$$y = 2 - 3(100) = -298$$

The point has $y = -302$, not -298, so it does not lie on the line.

FINAL ANSWER

$(9, -1.5)$, gradient -3, and no.

PAST PAPER QUESTION Q#: 2 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jan 2020 P1H - Jan 2020 | P1H

2 Find the lowest common multiple (LCM) of 28 and 105

....

(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

Write both numbers as products of prime factors:

$$28 = 2^2 \times 7$$

$$105 = 3 \times 5 \times 7$$

For the lowest common multiple, take the highest power of each prime:

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

FINAL ANSWER

420

PAST PAPER QUESTION

Q#: 3 **Marks: 4**

TOPIC: **Area & Perimeter**

Jan 2020 P1H - Jan 2020 | P1H

3 The diagram shows a shape.

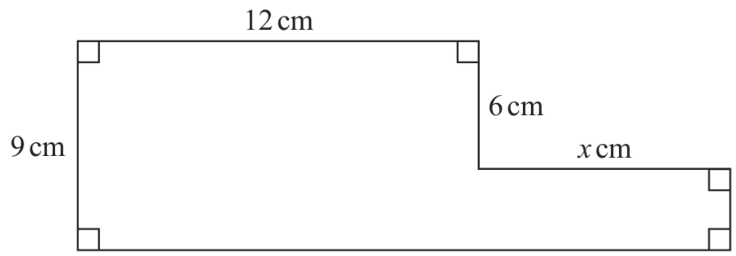

Diagram **NOT** accurately drawn

The shape has area 129 cm^2

Work out the value of x.

$x =$ ....

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Area & Perimeter**

Jan 2020 P1H - Jan 2020 | P1H

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Topic check. Area & Perimeter. The tag is correct.

Solution

Split the shape into two rectangles.
The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

The right rectangle has height

$$9 - 6 = 3 \text{ cm}$$

So its area is

$$3x$$

Total area:

$$108 + 3x = 129$$

$$3x = 21$$

$$x = 7$$

FINAL ANSWER
7 cm.

PAST PAPER QUESTION Q#: 4 Marks: 7

TOPIC: **Statistics Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	**Frequency**
$2 < w \leqslant 3$	12
$3 < w \leqslant 4$	16
$4 < w \leqslant 5$	9
$5 < w \leqslant 6$	2
$6 < w \leqslant 7$	1

(a) Write down the modal class.

....

(1)

(b) Work out an estimate for the mean weight of the 40 babies.

.... kg

(4)

One of the 40 babies is going to be chosen at random.

(c) Find the probability that this baby has a weight of more than 5 kg.

....

(2)

(Total for Question 4 is 7 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 7

TOPIC: **Statistics Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

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Topic check. Statistics Toolkit. The tag is correct.

Solution

(a) The greatest frequency is 16, so the modal class is

$$3 < w \leq 4$$

(b) Use class midpoints:

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

(c) Weights greater than 4 kg:

$$9 + 2 + 1 = 12$$

$$P(w > 4) = \frac{12}{40} = \frac{3}{10}$$

FINAL ANSWER

(a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

PAST PAPER QUESTION

TOPIC: **Ratio Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

Q#: 5 Marks: 6

5 120 children go on an activity holiday.
The ratio of the number of girls to the number of boys is 3 : 5

On Sunday, all the children either go sailing or go climbing.

$\frac{16}{25}$ of the boys go climbing.

Twice as many girls go sailing as go climbing.

Work out how many children go sailing on Sunday.

....

(Total for Question 5 is 6 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 6

TOPIC: **Ratio Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Ratio Toolkit. The tag is correct.

Method

Girls : boys = 3 : 5, so there are 8 parts.

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

$$\text{boys} = 120 \times \frac{5}{8} = 75$$

Boys climbing:

$$\frac{16}{25} \times 75 = 48$$

Boys sailing:

$$75 - 48 = 27$$

For the girls, twice as many go sailing as climbing, so the girls are split in the ratio

$$\text{sailing} : \text{climbing} = 2 : 1$$

Girls sailing:

$$\frac{2}{3} \times 45 = 30$$

Total sailing:

$$27 + 30 = 57$$

FINAL ANSWER

57

PAST PAPER QUESTION Q#: 6 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2020 P1H - Jan 2020 | P1H

6 (a) Write 7.8×10^{-4} as an ordinary number.

....

(1)

(b) Work out $\dfrac{5.6 \times 10^{4} + 7 \times 10^{3}}{2.8 \times 10^{-3}}$

Give your answer in standard form.

....

(2)

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$7.8 \times 10^{-4} = 0.00078$$

For part (b),

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Then

$$\frac{63000}{2.8 \times 10^{-3}} = \frac{63000}{0.0028}$$

$$= 22500000 = 2.25 \times 10^7$$

FINAL ANSWER

(a) 0.00078, (b) 2.25×10^7

PAST PAPER QUESTION

Q#: 7 Marks: 9

TOPIC: **Solving Linear Equations**

Jan 2020 P1H - Jan 2020 | P1H

7 (a) Expand and simplify $(m - 8)(m + 5)$

....

(2)

(b) Factorise fully $5y + 20y^2$

....

(2)

(c) Simplify $(p^2 + 3)^0$

....

(1)

(d) Solve $3(2x - 5) = \dfrac{9 - x}{2}$

Show clear algebraic working.

$x =$ ....

(4)

(Total for Question 7 is 9 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 9

TOPIC: **Solving Linear Equations**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Corrected to solving linear equations. The original tag was expanding brackets, but the largest single part is the equation.

Solution

(a) $(m-8)(m+5) = m^2 - 3m - 40$.
(b) $5y + 20y^2 = 5y(1+4y)$.
(c) $(p^2+3)^0 = 1$.
(d)

$$3(2x-5) = \frac{9-x}{2}$$

$$12x - 30 = 9 - x$$

$$13x = 39$$

FINAL ANSWER

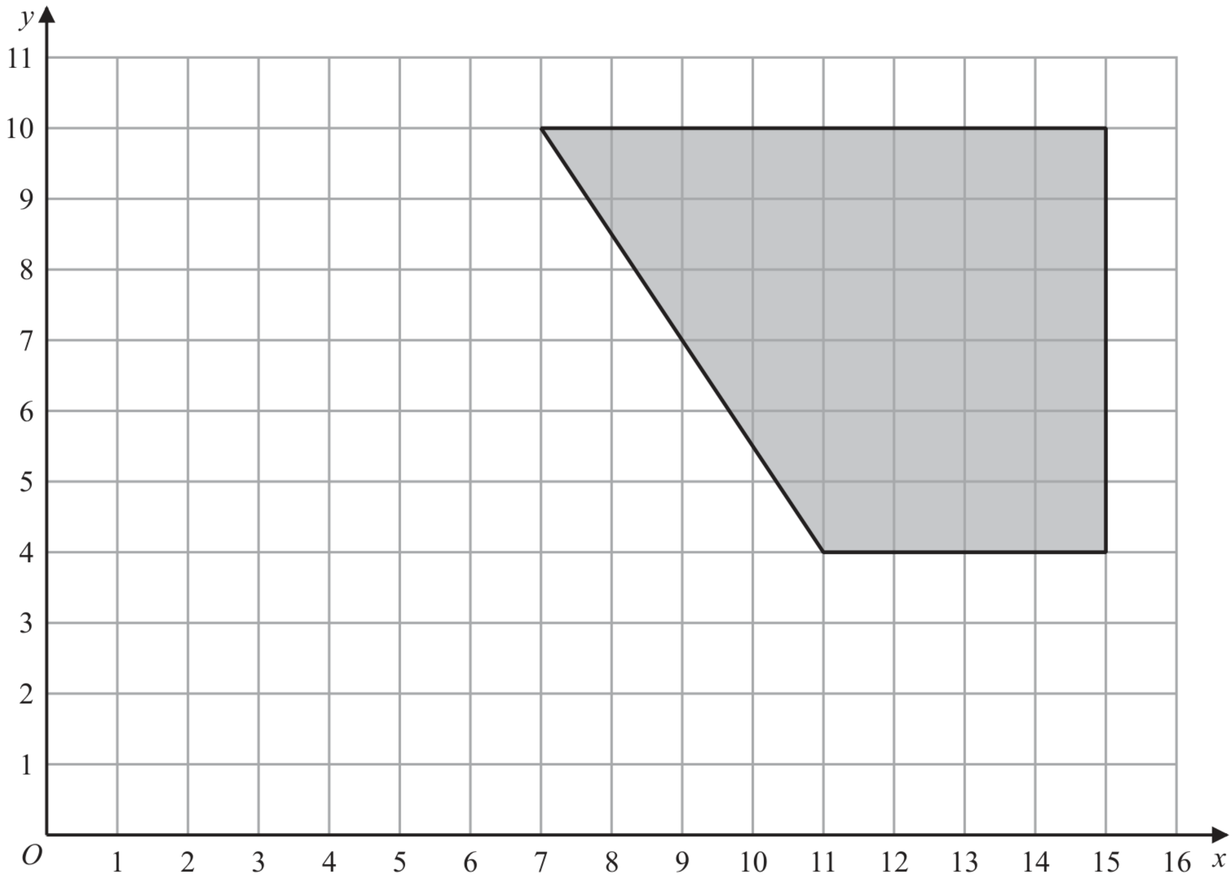
(a) $m^2 - 3m - 40$, (b) $5y(1+4y)$, (c) 1, (d) $x = 3$

PAST PAPER QUESTION Q#: 8 Marks: 2

TOPIC: **Transformations**

Jan 2020 P1H - Jan 2020 | P1H

8



On the grid, enlarge the shaded shape with scale factor $\frac{1}{2}$ and centre (1, 2)

(Total for Question 8 is 2 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 2

TOPIC: **Transformations**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Transformations. The tag is correct.

Solution

Use

$$\text{new point} = (1,2) + \frac{1}{2}(\text{old point} - (1,2))$$

The vertices

$$(7,10),\ (15,10),\ (15,4),\ (11,4)$$

become

$$(4,6),\ (8,6),\ (8,3),\ (6,3)$$

FINAL ANSWER

draw the enlarged shape with vertices $(4,6),(8,6),(8,3),(6,3)$.

PAST PAPER QUESTION

Topic: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

Q#: 9 **Marks: 3**

9 Here is a right-angled triangle.

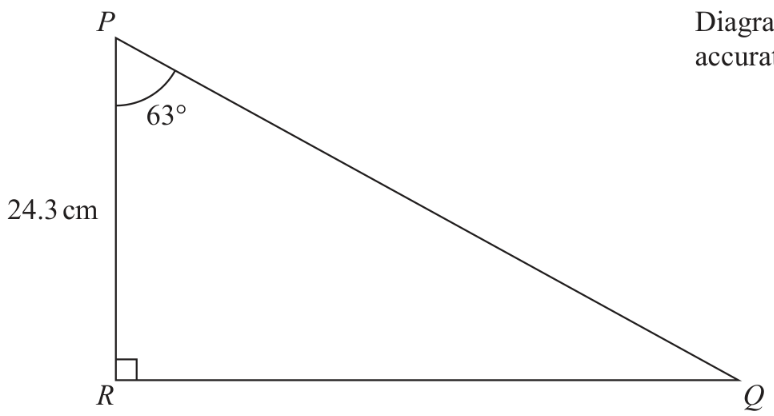

Diagram **NOT** accurately drawn

Calculate the length of PQ.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

$$\cos 63^\circ = \frac{24.3}{PQ}$$

$$PQ = \frac{24.3}{\cos 63^\circ} = 53.525\ldots$$

FINAL ANSWER

53.5 cm.

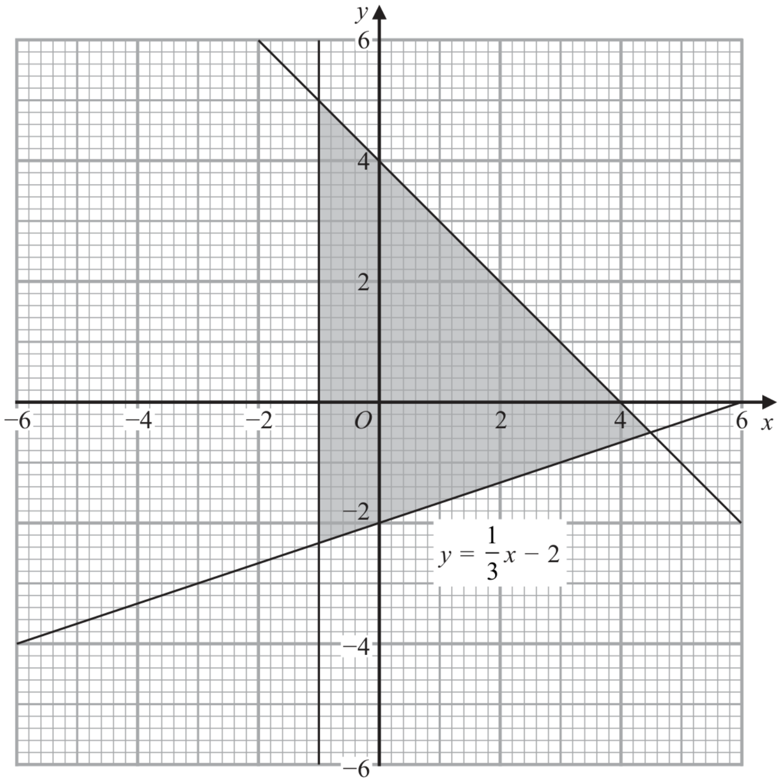
PAST PAPER QUESTION

Q#: 10 Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2020 P1H - Jan 2020 | P1H

10 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down the three inequalities that define the shaded region.

....

....

....

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2020 P1H - Jan 2020 | P1H

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Topic check. Correct. This is identifying inequalities from a shaded region.

Solution

The shaded region is left of the y-axis:

$$x \leq 0$$

It is below the line through $(0, 4)$ and $(4, 0)$:

$$y \leq -x + 4$$

It is above the line

$$y = \frac{1}{3}x - 2$$

So

$$y \geq \frac{1}{3}x - 2$$

FINAL ANSWER

$x \leq 0$, $y \leq -x + 4$, $y \geq \frac{1}{3}x - 2$.

PAST PAPER QUESTION Q#: 11 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jan 2020 P1H - Jan 2020 | P1H

11 Max invests \$6000 in a savings account for 3 years.
The account pays compound interest at a rate of 1.5% per year for the first 2 years.

The compound interest rate changes for the third year.
At the end of 3 years, there is a total of \$6311.16 in the account.

Work out the compound interest rate for the third year.
Give your answer correct to 1 decimal place.

.... %

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

After the first 2 years,

$$6000(1.015)^2 = 6181.35$$

Let the third-year interest rate be $r\%$.

$$6181.35\left(1 + \frac{r}{100}\right) = 6311.16$$

$$1 + \frac{r}{100} = \frac{6311.16}{6181.35} = 1.021000\ldots$$

$$r = 2.1000\ldots$$

FINAL ANSWER

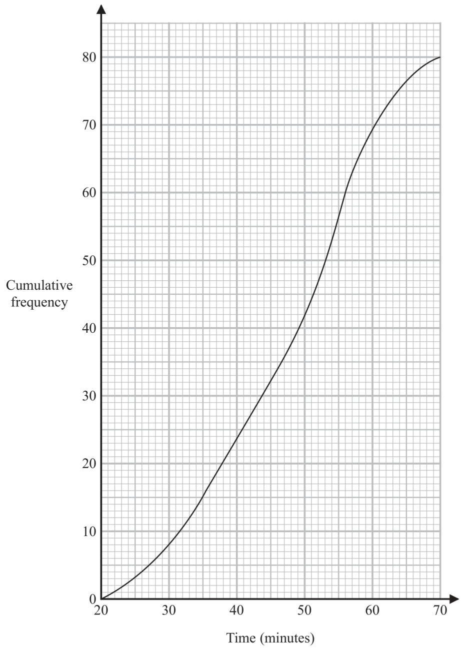
2.1%

PAST PAPER QUESTION Q#: 12 Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

Jan 2020 P1H - Jan 2020 | P1H

12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



(a) Use the graph to find an estimate for the interquartile range.

.... minutes
(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

(b) Work out an estimate for the number of men who took part in the race.

....
(3)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

There are 80 people.
For the interquartile range:

$$Q_1 \text{ is at cumulative frequency } 20$$

$$Q_3 \text{ is at cumulative frequency } 60$$

From the graph,

$$Q_1 \approx 38, \qquad Q_3 \approx 56$$

$$\text{IQR} \approx 56 - 38 = 18$$

At 50 minutes, the cumulative frequency is about 42.
No women took 50 minutes or less, so these 42 people are men.

$$0.6 \times \text{number of men} = 42$$

$$\text{number of men} = \frac{42}{0.6} = 70$$

FINAL ANSWER

IQR about 18 minutes, and about 70 men.

PAST PAPER QUESTION

Q#: 13 **Marks: 4**

TOPIC: **Standard & Compound Units**

Jan 2020 P1H - Jan 2020 | P1H

13 The diagram shows a solid cube.
The cube is placed on a table so that the whole of one face of the cube is in contact with the table.

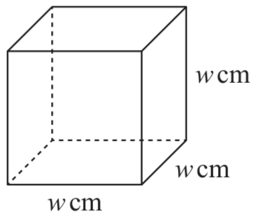

Diagram **NOT** accurately drawn

The cube exerts a force of 56 newtons on the table.
The pressure on the table due to the cube is 0.14 newtons/cm^2

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cube.

.... cm^3

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 4

TOPIC: **Standard & Compound Units**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Correct. This is a pressure and volume question.

Solution

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$0.14 = \frac{56}{\text{area}}$$

$$\text{area} = \frac{56}{0.14} = 400 \text{ cm}^2$$

The cube rests on one square face, so

$$w^2 = 400$$

$$w = 20$$

Volume of the cube is

$$20^3 = 8000$$

FINAL ANSWER

8000 cm^3.

PAST PAPER QUESTION

Q#: 14 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1H - Jan 2020 | P1H

14 The diagram shows parallelogram *EFGH*.

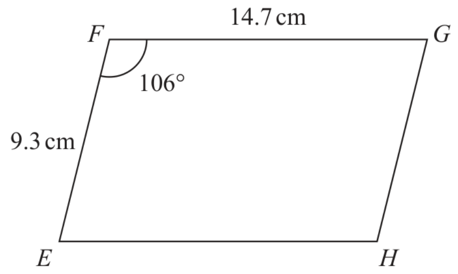

Diagram **NOT** accurately drawn

EF = 9.3 cm
FG = 14.7 cm
Angle *EFG* = 106°

(a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

.... cm^2
(2)

(b) Work out the length of the diagonal *EG* of the parallelogram.
Give your answer correct to 3 significant figures.

.... cm
(3)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Sine, Cosine Rule & Area of Triangles. The tag is correct.

Solution

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414\ldots$$

FINAL ANSWER
131 cm^2.

(b) In triangle EFG,

$$EG^2 = 9.3^2 + 14.7^2 - 2(9.3)(14.7)\cos 106^\circ$$

$$EG = 19.440\ldots$$

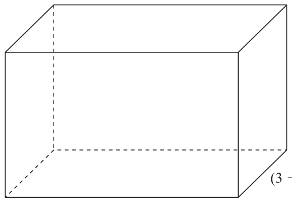
FINAL ANSWER
19.4 cm.

PAST PAPER QUESTION Q#: 15 Marks: 8

TOPIC: **Differentiation**

Jan 2020 P1H - Jan 2020 | P1H

15



The diagram shows a cuboid of volume V cm^3

(a) Show that $V = 15 + 16x - x^2 - 2x^3$

(3)

There is a value of x for which the volume of the cuboid is a maximum.

(b) Find this value of x.
Show your working clearly.
Give your answer correct to 3 significant figures.

$x =$ ....

(5)

(Total for Question 15 is 8 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 8

TOPIC: **Differentiation**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Correct. This is optimisation using differentiation.

Solution

The volume is

$$V = (2x+5)(x+1)(3-x)$$

First expand:

$$(2x+5)(x+1) = 2x^2 + 7x + 5$$

$$V = (2x^2 + 7x + 5)(3 - x)$$

$$V = 15 + 16x - x^2 - 2x^3$$

For the maximum volume,

$$\frac{dV}{dx} = 0$$

$$\frac{dV}{dx} = 16 - 2x - 6x^2$$

$$16 - 2x - 6x^2 = 0$$

$$3x^2 + x - 8 = 0$$

Using the quadratic formula,

$$x = \frac{-1 \pm \sqrt{97}}{6}$$

The valid positive value is

$$x = 1.4748\ldots$$

FINAL ANSWER

$x = 1.47$ to 3 significant figures.

PAST PAPER QUESTION

Q#: 16 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jan 2020 P1H - Jan 2020 | P1H

16 $P = \dfrac{2a - c}{d}$

$a = 58.4$ correct to 3 significant figures.
$c = 20$ correct to 2 significant figures.
$d = 3.6$ correct to 2 significant figures.

Work out the upper bound for the value of P.
Show your working clearly.
Give your answer correct to 2 decimal places.

....

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

$$P = \frac{2a - c}{d}$$

For the upper bound of P, make the numerator as large as possible and the denominator as small as possible.

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

$$P_{\text{upper}} = \frac{2(58.45) - 19.5}{3.55} = 27.4366\ldots$$

FINAL ANSWER

27.44

PAST PAPER QUESTION

Q#: 17 **Marks: 6**

Topic: **Algebraic Roots & Indices**

Jan 2020 P1H - Jan 2020 | P1H

17 (a) Show that $\left(6 + 2\sqrt{12}\right)^2 = 12\left(7 + 4\sqrt{3}\right)$

Show each stage of your working.

(3)

(b) Simplify fully $\left(\dfrac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

....

(3)

(Total for Question 17 is 6 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 6

TOPIC: **Algebraic Roots & Indices**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Corrected to Algebraic Roots & Indices. The question includes surds and fractional indices; the algebraic-index part is the stronger topic signal.

Method

First,

$$\sqrt{12} = 2\sqrt{3}$$

So

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Now square:

$$(6 + 4\sqrt{3})^2 = 36 + 48\sqrt{3} + 48$$

$$= 84 + 48\sqrt{3}$$

Factor out 12:

$$84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$$

For part (b),

$$\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}} = \left(\frac{t^{15}}{27a^{12}}\right)^{\frac{2}{3}}$$

$$= \frac{t^{10}}{9a^8}$$

FINAL ANSWER

(a) shown, (b) $\frac{t^{10}}{9a^8}$

PAST PAPER QUESTION

Q#: 18 Marks: 4

TOPIC: **Probability Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

18 There are 16 sweets in a bowl.

4 of the sweets are blackcurrant.
5 of the sweets are lemon.
7 of the sweets are orange.

Anna, Ravi and Sam each take at random one sweet from the bowl.

Work out the probability that the 5 lemon sweets are still in the bowl.

....

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Probability Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

There are 16 sweets and 5 are lemon, so 11 are not lemon.

$$P(\text{no lemon}) = \frac{11}{16} \times \frac{10}{15} \times \frac{9}{14}$$

$$= \frac{33}{112}$$

FINAL ANSWER

$\frac{33}{112}$.

PAST PAPER QUESTION

Q#: 19 Marks: 4

TOPIC: 3D Pythagoras & Trigonometry

Jan 2020 P1H - Jan 2020 | P1H

19 The diagram shows a cuboid *ABCDEFGH*.

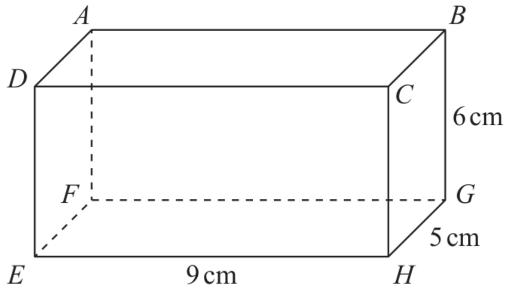

Diagram **NOT** accurately drawn

$EH = 9$ cm, $HG = 5$ cm and $GB = 6$ cm.

Work out the size of the angle between *AH* and the plane *EFGH*.
Give your answer correct to 3 significant figures.

.... °

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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Topic check. 3D Pythagoras & Trigonometry. The tag is correct.

Solution

The projection of AH onto the base plane is the diagonal of the base rectangle:

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

The vertical height is 6 cm, so

$$\tan\theta = \frac{6}{\sqrt{106}}$$

$$\theta = 30.232\ldots^\circ$$

FINAL ANSWER

30.2°.

PAST PAPER QUESTION Q#: 20 Marks: 4

TOPIC: **Graphs of Functions**

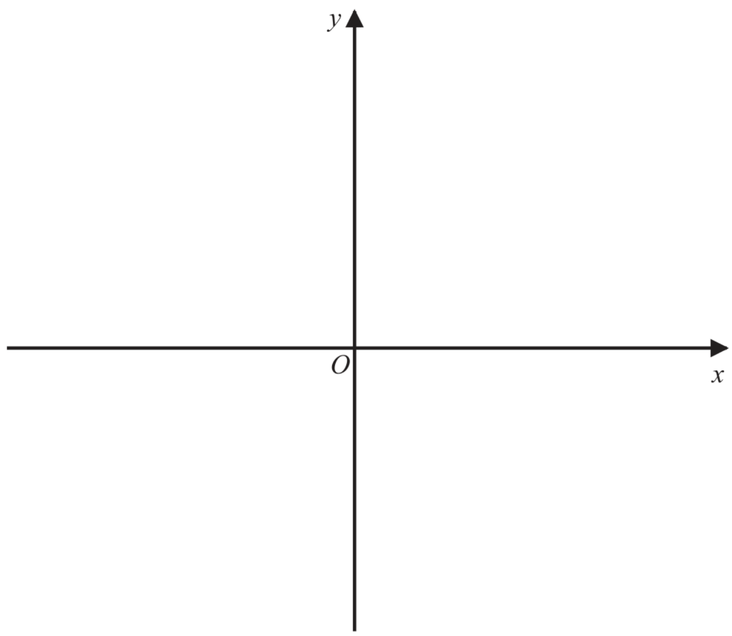
Jan 2020 P1H - Jan 2020 | P1H

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a,

(i) the coordinates of any points of intersection of **C** with the coordinate axes,

(ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Graphs of Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$y = 4(x-1)^2 - a$$

The turning point is read directly from the completed square form:

$$(1, -a)$$

For the y-intercept, put $x = 0$:

$$y = 4(0-1)^2 - a$$

$$y = 4 - a$$

So the y-intercept is

$$(0, 4-a)$$

For the x-intercepts, put $y = 0$:

$$4(x-1)^2 - a = 0$$

$$4(x-1)^2 = a$$

$$(x-1)^2 = \frac{a}{4}$$

$$x - 1 = \pm\frac{\sqrt{a}}{2}$$

$$x = 1 \pm \frac{\sqrt{a}}{2}$$

So the x-intercepts are

$$\left(1 - \frac{\sqrt{a}}{2}, 0\right) \quad \text{and} \quad \left(1 + \frac{\sqrt{a}}{2}, 0\right)$$

Since $a > 4$, the turning point is below the x-axis and the curve crosses the x-axis twice.

FINAL ANSWER

turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Graphs of Functions**

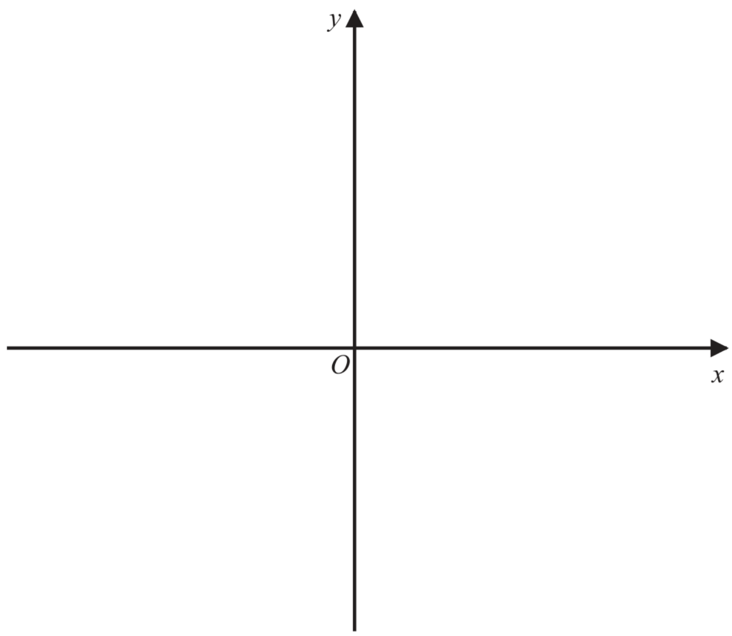
Jan 2020 P1H - Jan 2020 | P1H

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a,

(i) the coordinates of any points of intersection of **C** with the coordinate axes,

(ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

Topic: **Graphs of Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

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$$4(x-1)^2 = a$$

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Since $a > 4$, the turning point is below the x-axis and the curve crosses the x-axis twice.

FINAL ANSWER

turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

PAST PAPER QUESTION Q#: 21 Marks: 5

TOPIC: **Functions**

Jan 2020 P1H - Jan 2020 | P1H

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geqslant -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \ldots$

$h^{-1}(x) =$ ....

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$f(x) = x^2 - 2x$$

$$g(x) = x + 3$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = (x+3)^2 - 2(x+3)$$

$$h(x) = x^2 + 4x + 3$$

$$h(x) = (x+2)^2 - 1$$

Let

$$y = (x+2)^2 - 1$$

Since the domain is $x \geq -2$, take the positive square root:

$$x + 2 = \sqrt{y+1}$$

$$x = \sqrt{y+1} - 2$$

FINAL ANSWER

$h^{-1}(x) = \sqrt{x+1} - 2$

PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Functions**

Jan 2020 P1H - Jan 2020 | P1H

21 The functions f and g are such that

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$h^{-1}(x) =$ ....

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$f(x) = x^2 - 2x$$

$$g(x) = x + 3$$

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Since the domain is $x \geq -2$, take the positive square root:

$$x + 2 = \sqrt{y + 1}$$

$$x = \sqrt{y + 1} - 2$$

FINAL ANSWER

$h^{-1}(x) = \sqrt{x + 1} - 2$

PAST PAPER QUESTION

Topic: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

Q#: 22 Marks: 6

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$
J is the point with coordinates $(j, 15)$ where $j < 0$
K is the point with coordinates $(6, k)$

M is the midpoint of JK.
The gradient of HM is 2

Find the value of j and the value of k.

j = ...

k = ...

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 6

TOPIC: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

Since M is the midpoint of JK,

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2}\right)$$

The gradient of HM is 2, and $H = (-4, 1)$.

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

$$\frac{k+13}{j+14} = 2$$

$$k = 2j + 15$$

Also,

$$JK = \sqrt{80}$$

So

$$(6-j)^2 + (k-15)^2 = 80$$

Substitute $k = 2j + 15$:

$$(6-j)^2 + (2j)^2 = 80$$

$$j^2 - 12j + 36 + 4j^2 = 80$$

$$5j^2 - 12j - 44 = 0$$

$$(5j+10)(j-4.4) = 0$$

Since $j < 0$,

$$j = -2$$

Then

$$k = 2(-2) + 15 = 11$$

FINAL ANSWER

$j = -2,\ k = 11$

PAST PAPER QUESTION

Q#: 22 Marks: 6

TOPIC: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

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The gradient of HM is 2

Find the value of j and the value of k.

$j =$ ....

$k =$ ....

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

TOPIC: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

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Method

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Since $j < 0$,

$$j = -2$$

Then

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FINAL ANSWER

$j = -2, \ k = 11$